

Proposition 1 (The Implicit Function Theorem). *(What is F?)* $F : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, If $F(a, b) = 0$ and $F(x, y)$ is continuously differentiable on some open disk with center (a, b) , then, if

$$\frac{\partial F}{\partial y}(a, b) \neq 0$$

there exists an $h > 0$ and a unique function $\phi(x)$ continuously differentiable *(What can we say about ϕ ?)* defined for $|x - a| < h$ such that $\phi(a) = b$ and $F(x, \phi(x)) = 0$ for $|x - a| < h$.

Proposition 2 (The Actual Pasting Lemma). *Let X and Y be metric spaces. Let $(A, B) \in \text{Closed}(X)^2$ and assume further that $X = A \cup B$. Assume that we have two functions $f : A \rightarrow Y$ and $g : B \rightarrow Y$. Assume that $f(x) = g(x) \quad \forall x \in A \cap B$. Then we can define a new function $h : X \rightarrow Y$ by:*

$$h(x) = \begin{cases} f(x) & \text{if } x \in A, \\ g(x) & \text{if } x \in B \end{cases}$$

*(if $x \in A \cap B$ then we have no ambiguity by assumption). **Claim:** If both f and g are continuous, then h is continuous.*

Definition 1. Let $P \in \mathbb{R}[x, y]$ be a squarefree polynomial, with $\deg_y(P) > 0$. We denote by $\text{proj}(P)$ the set of points $x \in \mathbb{R}$ such that either $\text{Disc}_y(P)(x) = 0$ or $\text{lc}_y(P)(x) = 0$.

Exercise 1. Let $P \in \mathbb{R}[x, y]$ be a squarefree polynomial, with $\deg_y(P) > 0$. Suppose that $\#\text{proj}(P) < \infty$. Let $x_1, x_2, \dots, x_s$ be the elements of $\text{proj}(P)$ in ascending order, and let $x_0 = -\infty$ and $x_{s+1} = +\infty$. Show that for all $i = 0, 1, \dots, s$, the following properties hold:

1. For all $x \in (x_i, x_{i+1})$, $\{y \in \mathbb{R} : P(x, y) = 0\}$ is finite.
2. For all $(x, y) \in (x_i, x_{i+1}) \times \mathbb{R}$ such that $P(x, y) = 0$, there exists an open neighborhood I_x of x and a unique function $g \in \mathcal{C}^1(I_x)$ such that $g(x) = y$ and $\forall x' \in I_x$, $P(x', g(x')) = 0$.

Let $x \in (x_i, x_{i+1})$, $t = \#\{y \in \mathbb{R} : P(x, y) = 0\}$, and let $y_1, y_2, \dots, y_t$ be the roots of $P(x, y)$ in ascending order. Show that the following properties hold:

3. For all $j = 1, 2, \dots, t$, let g_j be the local function given by the previous lemma at (x, y_j) . Show that g_j extends uniquely to a continuous function $h_j : (x_i, x_{i+1}) \rightarrow \mathbb{R}$ such that $h_j(x) = y_j$ and for all $x' \in (x_i, x_{i+1})$, $P(x', h_j(x')) = 0$.
4. In addition, $h_1(x') < h_2(x') < \dots < h_t(x')$ for all $x' \in (x_i, x_{i+1})$.
5. For all $x' \in (x_i, x_{i+1})$, $\#\{y \in \mathbb{R} : P(x', y) = 0\} \geq t$.

Finally, show that the number of roots of $P(x, y)$ in $\mathbb{R}$ is constant for all $x \in (x_i, x_{i+1})$.

Proof. 1. For property 1, We know that $\forall x \in (x_i, x_{i+1})$, $lc_y(P)(x) \neq 0$, so the polynomial in y $P(x, y)$ isn't the zero polynomial. It's impossible that $P(x, y)$ has an infinite number of roots. In conclusion, $\{y \in \mathbb{R} : P(x, y) = 0\}$ is finite.

2. For property 2, $\forall (x, y) \in (x_i, x_{i+1}) \times \mathbb{R}$, such that $P(x, y) = 0$, we have $Disc_y(P)(x) \neq 0$, so $P(x, Y)$ don't have multiple roots, and $P(x, y) = 0$, so $\frac{\partial P}{\partial y}(x, y) \neq 0$. P is polynomial function, so it's continuously differentiable. According to the Implicit Function Theorem, there exists an open neighborhood I_x of x , and a unique function $g \in \mathcal{C}^1(I_x)$, such that $g(x) = y$ and $\forall x' \in I_x$, $P(x', g(x')) = 0$.

3. For property 3, Fix $j \in \{1, 2, \dots, t\}$. By property 2, at the point (x, y_j) there is an open interval $I_0 \subset (x_i, x_{i+1})$ containing x , and there is a function $g_j : I_0 \rightarrow \mathbb{R}$ such that $g_j(x) = y_j$ and $P(u, g_j(u)) = 0$ for all $u \in I_0$. We want to extend g_j as far as possible. Let $h : I \rightarrow \mathbb{R}$ be a maximal extension of g_j , where $I \subset (x_i, x_{i+1})$ is an open Interval containing x . This means that $h(u) = g_j(u)$ for all $u \in I \cap I_0$, and $P(u, h(u)) = 0$ for all $u \in I$. Also, I is maximal with this property. We will prove that $I = (x_i, x_{i+1})$. Suppose not. We first consider the right endpoint of I . Write $I = (\alpha, \beta)$. Assume, for contradiction, that $\beta < x_{i+1}$. Then $\beta \in (x_i, x_{i+1})$. Since there is no projection point in (x_i, x_{i+1}) , we have $lc_y(P)(\beta) \neq 0$ and $Disc_y(P)(\beta) \neq 0$. Write P as a polynomial in y : $P(u, y) = a_d(u)y^d + a_{d-1}(u)y^{d-1} + \dots + a_0(u)$, where $a_d(u) = lc_y(P)(u)$. Since $a_d(\beta) \neq 0$, there is a small open interval U containing β , and there is a constant $c > 0$, such that $|a_d(u)| \geq c$ for all $u \in U$. Also, the functions $a_0(u), a_1(u), \dots, a_{d-1}(u)$ are continuous, so they are bounded on a smaller interval around β . Therefore, the roots of $P(u, Y)$ are uniformly bounded for u close to β . Since $P(u, h(u)) = 0$ for all $u \in I$, this shows that $h(u)$ is bounded when u is close to β from the left. Now choose a sequence $u_n \in I$ such that $u_n \rightarrow \beta$ and $u_n < \beta$. Since $h(u_n)$ is bounded, we may take a subsequence such that $h(u_n) \rightarrow y_*$ for some $y_* \in \mathbb{R}$. For simplicity, we still write this subsequence as u_n . For every n , we have $P(u_n, h(u_n)) = 0$. Taking the limit and using the continuity of P , we get $P(\beta, y_*) = 0$. Since $Disc_y(P)(\beta) \neq 0$, the polynomial $P(\beta, Y)$ has no multiple root. Hence y_* is a simple root of $P(\beta, Y)$. Therefore $\frac{\partial P}{\partial y}(\beta, y_*) \neq 0$. By the implicit function theorem, there is an open interval K containing β , and there is a function $\varphi : K \rightarrow \mathbb{R}$ such that $\varphi(\beta) = y_*$ and $P(u, \varphi(u)) = 0$ for all $u \in K$. We claim that h and φ are equal on a small interval to the left of β . Indeed, since $(u_n, h(u_n)) \rightarrow (\beta, y_*)$, the points $(u_n, h(u_n))$ are eventually inside the neighborhood where the implicit function theorem gives a unique local branch. Hence, for all sufficiently large n , $h(u_n) = \varphi(u_n)$. By the local uniqueness in the implicit function theorem, this equality extends to a small interval around such a point. Thus there exists $\varepsilon > 0$ such that $h(u) = \varphi(u)$ for all $u \in (\beta - \varepsilon, \beta)$. Now by the pasting lemma we can define a new function $\tilde{h} : I \cup K \rightarrow \mathbb{R}$

by $\tilde{h}(u) = \begin{cases} h(u), & u \in I, \\ \varphi(u), & u \in K. \end{cases}$ This definition is well-defined because h and

φ are equal on the overlap near β . Also, $P(u, \tilde{h}(u)) = 0$ for all $u \in I \cup K$. But K contains points larger than β , so $I \cup K$ is strictly larger than I . This contradicts the maximality of I . Therefore the right endpoint of I cannot lie inside (x_i, x_{i+1}) . So we must have $\beta = x_{i+1}$ if x_{i+1} is finite, and $\beta = +\infty$ if $x_{i+1} = +\infty$. The same argument at the left endpoint shows that the left endpoint of I cannot lie inside (x_i, x_{i+1}) . Hence $I = (x_i, x_{i+1})$. Therefore g_j extends to a continuous function $h_j : (x_i, x_{i+1}) \rightarrow \mathbb{R}$ such that $h_j(x) = y_j$ and $P(u, h_j(u)) = 0$ for all $u \in (x_i, x_{i+1})$. It remains to prove uniqueness. Suppose that $\tilde{h}_j : (x_i, x_{i+1}) \rightarrow \mathbb{R}$ is another continuous extension of g_j such that $P(u, \tilde{h}_j(u)) = 0$ for all $u \in (x_i, x_{i+1})$. Define $A = \{u \in (x_i, x_{i+1}) : h_j(u) = \tilde{h}_j(u)\}$. Since both functions extend g_j , the set A is nonempty. Also, since h_j and $\tilde{h}_j$ are continuous, A is closed. Now let $u_0 \in A$, and set $v_0 = h_j(u_0) = \tilde{h}_j(u_0)$. Since $u_0 \in (x_i, x_{i+1})$, we have $\text{Disc}_y(P)(u_0) \neq 0$. Thus $\frac{\partial P}{\partial y}(u_0, v_0) \neq 0$. By the local uniqueness in the implicit function theorem, h_j and $\tilde{h}_j$ are equal in a small neighborhood of u_0 . Hence A is open. So A is nonempty, open and closed in the interval (x_i, x_{i+1}) . Since this interval is connected, we get $A = (x_i, x_{i+1})$. Therefore $h_j = \tilde{h}_j$. This proves uniqueness.

4. For property 4, suppose that there exists an index $x' \in (x_i, x_{i+1})$, such that there exists h_{j1}, h_{j2} and $h_{j1}(x') > h_{j2}(x')$, and $h_{j1}(x') = y_1 < h_{j2}(x') = y_2$, so $h_{j1}(x') = h_{j2}(x')$, so x' is a root multiple, so $\text{Disc}_y(x') \neq 0$. But $x' \in (x_i, x_{i+1})$, contradiction, so $\forall x' \in (x_i, x_{i+1}), h_1(x') < h_2(x') < \dots < h_t(x')$
5. For property 5, Let $x' \in (x_i, x_{i+1})$ be fixed. By property 3, for each $j = 1, 2, \dots, t$, there exists a continuous function $h_j : (x_i, x_{i+1}) \rightarrow \mathbb{R}$ such that $P(x', h_j(x')) = 0$. Therefore $h_1(x'), h_2(x'), \dots, h_t(x')$ are real roots of the polynomial $P(x', Y)$. By property 4, we have $h_1(x') < h_2(x') < \dots < h_t(x')$. Thus these t roots are pairwise different. Hence the polynomial $P(x', Y)$ has at least t distinct real roots. Therefore $\#\{y \in \mathbb{R} : P(x', y) = 0\} \geq t$. Since x' was arbitrary, the result holds for all $x' \in (x_i, x_{i+1})$.
6. Take two arbitrary points $a, b \in (x_i, x_{i+1})$. Let $t_a = \#\{y \in \mathbb{R} : P(a, y) = 0\}$ and $t_b = \#\{y \in \mathbb{R} : P(b, y) = 0\}$. First, apply property 5 with the initial point a . Then for every point in (x_i, x_{i+1}) , the number of real roots is at least t_a . In particular, at the point b , we get $t_b \geq t_a$. Next, apply property 5 with the initial point b . Then for every point in (x_i, x_{i+1}) , the number of real roots is at least t_b . In particular, at the point a , we get $t_a \geq t_b$. Hence $t_a = t_b$. Since a and b were arbitrary, the number $\#\{y \in \mathbb{R} : P(x, y) = 0\}$ is the same for all $x \in (x_i, x_{i+1})$. Therefore the number of real roots of $P(x, Y)$ is constant on (x_i, x_{i+1}) . □